

PURPOSE

- Detect **doomscrolling behavior** from YouTube user activity
- Build **hybrid ML system** for behavioral classification
- Encourage **mindful browsing** through subtle intervention nudges

BACKGROUND

- Users consume **endless short-form content** with **low engagement**
- Rapid scrolling reduces reflection and increases **unintentional use**
- Driven by **recommendation systems** and **continuous feeds**
- Leads to **reduced focus** and **lower productivity**
- Difficult to detect since it depends on behavior, not time
- Common in **short-form platforms** like YouTube Shorts
- Existing Solutions
 - **Screen time limits** restrict usage but ignore behavior
 - **App blockers** disrupt both harmful and intentional use
- Lack **real-time detection** of user interaction patterns

Data Sources & Signals

- Manually labeled **YouTube sessions** as neutral vs doomscrolling
- Collected **25 neutral** and **25 doomscrolling sessions** across feeds
- Included sessions from **short-form** and **long-form** YouTube content
- Extracted **behavioral signals**: scroll rate, clicks, watch time, session duration
- Used dataset for **hybrid ML classification training** and **evaluation**

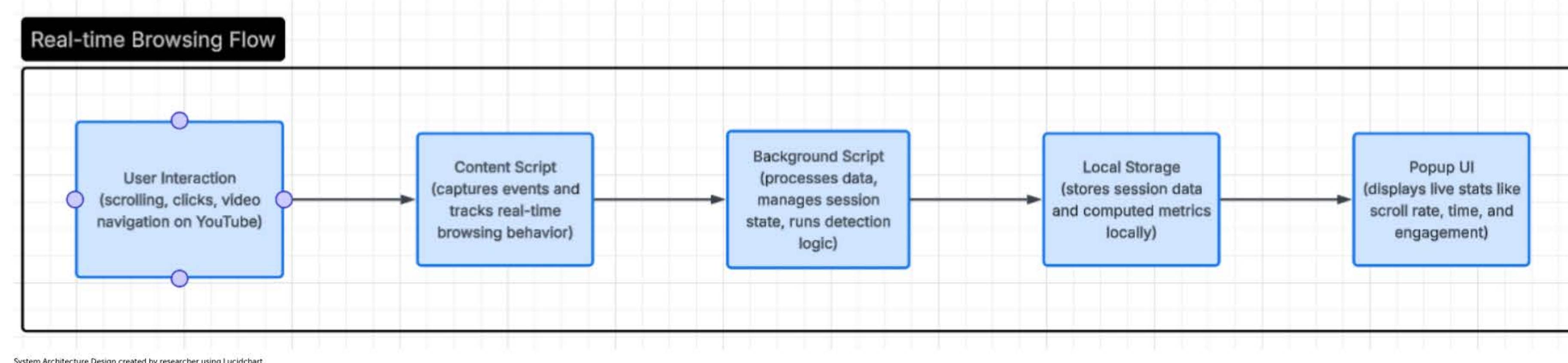


ML-Driven Browser Extension to Reduce Content Scrolling

METHODOLOGY

System Architecture

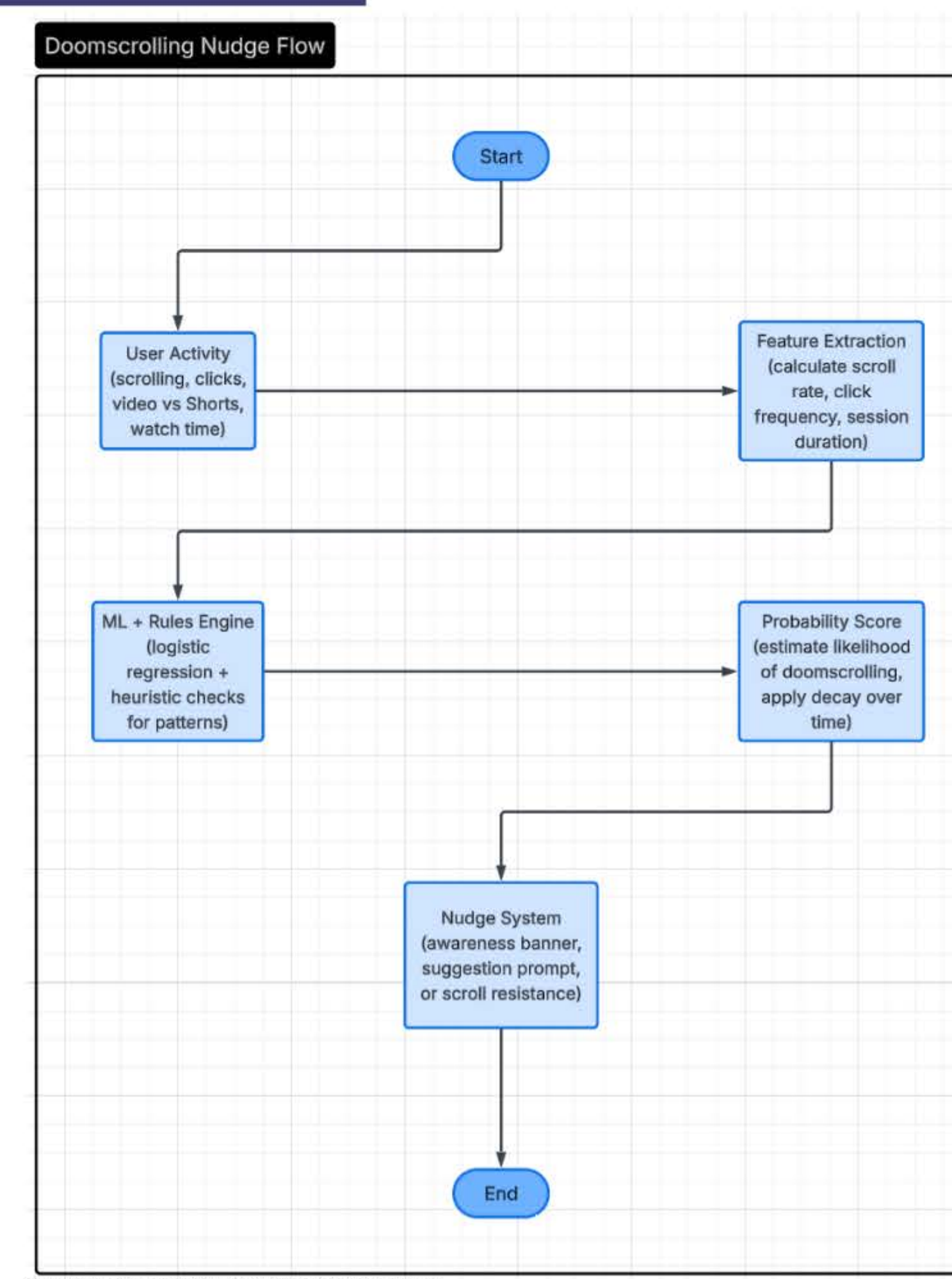
- Built a **Firefox extension** using content, background, and popup scripts to monitor browsing activity in real time
- Content script captures **scrolling, clicks, and video interaction data**, while background script manages session processing and storage
- Applied **event filtering techniques** such as throttling and deduplication to reduce noise and improve signal quality
- Popup interface displays **live session metrics** including scroll rate, watch time, and engagement summaries



METHODOLOGY [CONT.]

Detection + ML System

- Extracted **behavioral features** including scroll rate, click frequency, session duration, and video vs Shorts interaction patterns
- Implemented a **hybrid detection model** combining rule-based heuristics with a logistic regression classifier trained on labeled sessions
- Used **probability scoring with decay functions** to prevent saturation and maintain stable long-session predictions
- Triggered **multi-level nudges** based on confidence thresholds to encourage user awareness without blocking content



RESULTS

Design Goals:

- Captured real-time **YouTube scrolling, clicking, and watch behavior** across sessions.
- Hybrid detection system differentiated **doomscrolling behavior from neutral browsing patterns**.
- ML classification model produced stable probability outputs across varying session contexts.
- Three-tier intervention system (awareness, suggestions, resistance) activated based on engagement signals.
- Fully browser-based implementation enabled **real-time processing without external computation or data loss**.

Limitations

- Dataset limited to 50 manually labeled sessions, restricting generalization and large-scale validation.
- Model evaluation lacks **external benchmarking** and **multi-user testing**, limiting statistical confidence.

CONCLUSION

- Developed an **ML-driven browser extension** that detects doomscrolling behavior in real time on YouTube.
- Combined **behavioral signals, rule-based logic, and machine learning** into a unified detection framework.
- Produced **context-aware interventions** that encourage mindful browsing without blocking content.
- Demonstrates potential for **AI-based** digital wellness tools using lightweight in-browser systems.

REFERENCES

API Reference | YouTube Data API. (n.d.). Google Developers. <https://developers.google.com/youtube/v3/docs>

Bella, F., & Silvestri, M. (2017). Effects of directional auditory and visual warnings at intersections on reaction times and speed reduction times. Transportation Research Part F: Traffic Psychology and Behaviour, 51, 88–102. <https://doi.org/10.1016/j.trf.2017.09.006>

Browser Extensions - Mozilla | MDN. (n.d.). Developer.mozilla.org. <https://developer.mozilla.org/en-US/docs/Mozilla/Add-ons/WebExtensions>

Tools Used

Lucidchart — system diagrams
Firefox WebExtensions API — extension framework
JavaScript and HTML — core implementation
Browser Developer Tools — debugging and testing
Local Storage API — session data storage